

Figure 5: A simple plot with Matplotlib

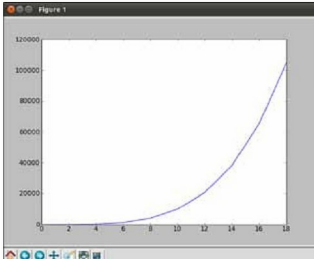


Figure 6: The plot after utilising the arange function

```
13
14 In[4]: z = np.array([[0,0],[0,1]])
15
16 In[5]: np.linalg.det(z)
17
18 Out[5]: 0 #z is a singular matrix and hence has its
determinant as zero
```

Integration

The `scipy.integrate` package provides several integration techniques, which can be used to solve simple and complex integrations. The package provides various methods to integrate functions. We will be discussing a few of them here. Let us first understand how to integrate the following functions:

$$\int_0^9 x^2 dx$$

$$\int_1^{2\sqrt{x}} \frac{1}{\sqrt{x}} dx$$

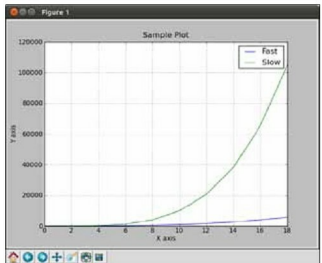


Figure 7: A multi-line plot with Matplotlib

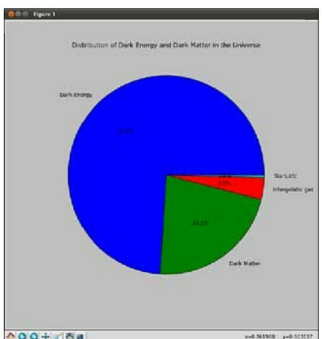


Figure 8: A pie chart with Matplotlib

```
# Simple Integration of x^2

In[1]: from scipy.integrate import quad

In[2]: import scipy as sp

In[3]: sp.integrate.quad(lambda x: x**2,0,3)

Out[3]: (9.0, 9.9922072216264089e-14)

# Integration of 2*sqrt(x)/sqrt(x)
```